

Satish Kumar

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Professional Summary:

Accomplished Senior Cloud/ DevOps Engineer with 11 years of experience in designing, automating, and managing cloud infrastructures across AWS, Azure, GCP and on-premises environments. Proficient in Infrastructure such as Code (Terraform, CloudFormation, Ansible), CI/CD pipelines (Jenkins, GitHub Actions, Spinnaker), and containerization/orchestration (Docker, Kubernetes, Helm). Skilled in scripting with Python, Bash, and PowerShell, and experienced in monitoring/log management tools such as Prometheus, Grafana, ELK Stack, Splunk, and Datadog. Experienced in supporting AI-enabled and data-driven applications by building secure, scalable platforms, with a focus on automation, reliability, and compliance in enterprise environments.

- Proficient in managing cloud services, including AWS (EC2, S3, RDS, Lambda, EKS) and Azure (AKS, Service Bus, Functions, Storage, Cosmos DB).
- Extensive experience in Infrastructure as Code (IaC) using Terraform, CloudFormation, and Ansible for provisioning, automation, and configuration management.
- Skilled in implementing CI/CD pipelines using tools such as Jenkins, GitHub Actions, Ansible, Spinnaker, and Azure DevOps to enable seamless build, test, and deployment processes.
- Expertise in containerization and orchestration technologies, including Docker, Kubernetes, and Helm, for efficient microservices deployment and scaling.
- Proficient in Azure DevOps (ADO), Terraform, Ansible, and Docker, enabling seamless deployment, configuration management, and automation for cloud-native applications.
- Strong knowledge of monitoring and log management tools like Prometheus, Grafana, ELK Stack, Splunk, Datadog, and New Relic for observability and troubleshooting.
- Supported AI and GenAI-based solutions by building secure, scalable platforms and deployment pipelines in regulated environments.
- Expertise in configuring Kubernetes clusters, deploying applications using YAML manifests, and managing AKS/EKS/GKE for scalable infrastructures.
- Proficient in managing databases, including PostgreSQL, MongoDB, Cassandra, DynamoDB, and Azure SQL, with expertise in schema design and optimization.
- Experienced DevOps Engineer with expertise in Google Cloud Platform (GCP), specializing in cloud infrastructure automation, CI/CD pipelines, and Kubernetes orchestration to enhance scalability, security, and efficiency.
- Hands-on experience in configuring and managing CI/CD pipelines integrated with SonarQube and Veracode for static code analysis and security testing.
- Extensive experience with source control management tools like Git, GitHub, Bitbucket, and Subversion for version control and collaboration.
- Strong expertise in managing load balancers, Auto Scaling, Route 53 DNS, and network configurations (VPCs, subnets, security groups) for highly available systems.
- Proficient in building event-driven architectures using AWS Lambda and Kinesis for real-time data processing.
- Implemented optimized deployment strategies using Spinnaker, AKAMAI, and automated rollbacks to ensure zero-downtime releases.
- Experienced in managing application and web servers like Tomcat, JBoss, and WebSphere, including automation of deployments with Ansible and Jenkins.
- Skilled in using monitoring tools like Datadog and OpenStack utilities for tracking microservices performance on Kubernetes.
- Integrated Kafka for real-time traffic analysis and monitored upstream/downstream metrics using Grafana dashboards.
- Expertise in using Apache Tomcat, Maven, and Nexus for Java application deployment and build management.
- Demonstrated strong project management skills, participating in SCRUM ceremonies and managing JIRA/Confluence tools for efficient team collaboration.

Certifications:

- AWS Certified DevOps Engineer – Professional
 - GCP – Professional Cloud Architect
 - Microsoft Certified: DevOps Engineer - Expert
 - AWS Certified Developer – Associate
 - Microsoft Certified: Azure Administrator – Associate
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Technical Skills:

Cloud Platforms & Services:

AWS: EC2, S3, VPC, EBS, EFS, RDS, DynamoDB, Lambda, CloudWatch, Auto Scaling, Route 53, ELB, CloudFormation, ECS, ECR, Kinesis, SNS, SQS, SSM, Connect, CodeDeploy.

Azure: Blob Storage, Files, Functions, Service Bus, Key Vault, AD, API Management, SQL/DB, Cosmos DB, Monitor, Log Analytics, OpenStack.

Containerization &

Orchestration:

Docker, Kubernetes, Helm, KOPS (Kubernetes Operations).

CI/CD & Build Tools:

Jenkins, Maven, Spinnaker, Nexus, Artifactory, uDeploy, Azure DevOps.

Infrastructure as Code &

Configuration Management:

Terraform, AWS CloudFormation, Ansible, Chef, Puppet.

Version Control &

Repository Management:

Git, GitHub, Bitbucket, Subversion (SVN).

Monitoring & Logging:

ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus & Grafana, Splunk, Datadog, New Relic, Azure Application Insights, AWS CloudWatch.

Programming &

Scripting Languages:

Python, Ruby, PowerShell, Bash/Shell scripting, Perl.

Web Servers &

Application Servers:

Apache Tomcat, JBoss, WebSphere, IIS, TC Server.

Databases:

PostgreSQL, MongoDB, Cassandra, DynamoDB, Azure SQL, Cosmos DB.

Message Queuing & Streaming:

Apache Kafka, AWS SQS, Azure Service Bus.

Security & Authentication:

AWS IAM, Multi-Factor Authentication (MFA), Security Groups, Network ACLs, Azure AD, Azure Key Vault.

Project Management &

Collaboration:

JIRA, Confluence, Agile/SCRUM methodology.

Operating Systems:

Linux (RHEL, CentOS, Ubuntu), Windows Server.

Quality Assurance & Testing:

SonarQube, Veracode.

Network & Content Delivery:

Akamai, Load Balancers, DNS management.

Professional Experience:

Client: CVS Health, Houston, TX.

Mar 2021 – Present

Role: Sr Cloud Engineer

Responsibilities:

- Managed AWS infrastructure across EC2, S3, EFS, RDS, Lambda, Kinesis, ECS, ECR, CloudWatch, and VPC services to support enterprise applications across multiple environments.
- Designed, configured, and supported Amazon Connect contact flows, routing profiles, Lambda integrations, and real-time agent metrics for enterprise-scale call centers.
- Contributed to responsible AI adoption by aligning infrastructure and release processes with enterprise approval, governance, and audit expectations.
- Automated application server and Jenkins node lifecycle management using Ansible playbooks, ensuring seamless rehydration of servers without losing data or configurations.
- Supported GenAI-enabled member service initiatives by building secure, scalable AWS infrastructure for conversational AI and AI-assisted support workflows in a regulated healthcare environment.

- Automated cloud deployments using Ansible, Python, AWS CloudFormation, and Terraform, reducing manual intervention and streamlining the deployment process.
- Administered AWS services including IAM (user/role/policy management), VPC, ELB, EFS, EC2, S3, and RDS
- Built deployment guardrails, logging, secrets management, and environment standards for AI-enabled applications, helping teams meet security, compliance, and operational requirements.
- Analyzed time-series metric logs for upstream and downstream traffic using Prometheus/Kafka and Grafana Dashboards.
- Performed OS and application patching across AWS and on-prem environments, ensuring compliance and security standards
- Automated infrastructure creation, deployment, and recovery using CloudFormation, Docker, and Jenkins in both cloud and on-premises environments, ensuring disaster recovery capabilities.
- Managed and administered VMware vSphere environments, including VM provisioning, clustering, and resource optimization
- Containerized microservices using Docker, utilizing Docker files and Docker Compose for cost optimization and efficient management of containerized applications.
- Automated Kubernetes cluster creation using AWS, Ansible, and KOPS, storing configuration files in S3 for centralized management.
- Supported Nutanix clusters (AHV), handling storage, compute, and virtualization lifecycle operations
- Monitored system health and performance using tools like CloudWatch, Prometheus, and Grafana, ensuring high availability
- Designed and implemented disaster recovery plans using Python, ensuring business continuity and minimizing downtime during critical events.
- Managed Linux servers including RHEL 8/9, Amazon Linux, Oracle Linux, and Ubuntu 22.04
- Developed event-driven architectures using AWS Lambda for real-time data processing and improved application responsiveness.
- Managed the creation and deployment of Kubernetes clusters, including pods, replica sets, services, and deployments via YAML files.
- Configured, maintained, and customized JIRA and Confluence for better team coordination and streamlined ticketing system management.

Environment: Python, Terraform, AWS, EC2, Security Groups, S3, EBS, EFS, SQS, SNS, RDS, ASG, DNS, CloudWatch, Lambda, Kinesis, SSM, CodeDeploy, Connect, ELK, ECS, ECR, VPC, Jenkins, Ansible, PostgreSQL, RESTful APIs, Nexus Docker, CI/CD, IaC, GitHub, Spinnaker, Akamai, Tomcat, Maven, PowerShell, EC2 Auto Scaling, KOPS, Kubernetes, DMS, YUM, SSM, JIRA, Confluence, SCRUM, Kafka, Datadog, ELK Stack.

Client: Applied Materials, San Jose, CA.

Aug 2018 – Jan 2021

Role: Sr Cloud Engineer

Responsibilities:

- Designed and implemented a continuous integration system using Jenkins by creating Python and Shell scripts to streamline build and deployment processes.
- Developed Python, Perl, and shell/bash scripts for TC Server/Tomcat web application servers and automated the build and release process and certain processes for the smooth process of building tools like Maven.
- Automated infrastructure management using Infrastructure as Code (IaC) tools such as Terraform and Google Deployment Manager, reducing deployment times by 40%.
- Performed VM migrations, snapshots, and capacity planning across virtualized infrastructure
- Managed Windows services, IIS configurations, and OS-level troubleshooting in enterprise environments
- Designed and deployed scalable, secure infrastructure on Google Cloud Platform (GCP) utilizing services such as Compute Engine, Cloud Storage, Cloud SQL, and VPC networking to support microservices architecture.
- Built and supported GCP infrastructure for conversational AI workloads used in member-facing healthcare experiences, with a focus on secure deployment, scalability, and operational reliability.

- Implemented and managed Kubernetes clusters (GKE) on GCP, ensuring efficient deployment, scaling, and management of containerized applications in a microservices environment.
- Partnered with application and product teams to enable GenAI-based member support solutions on Google Cloud, using Cloud Run, GKE, Pub/Sub, and BigQuery for integration, data flow, and observability.
- Configured and optimized CI/CD pipelines using Jenkins integrated with GCP services to automate build, test, and deployment processes for microservices.
- Applied AI/ML techniques to automation workflows for predictive scaling, intelligent alerting, and anomaly detection in cloud environments.
- Implemented platform controls for GenAI services, including IAM hardening, secret management, audit logging, and environment-level guardrails to support responsible AI adoption in a regulated healthcare setting.
- Integrated monitoring tools like Prometheus, Grafana, and OpenStack's built-in utilities to track resource usage, improving observability and proactive system maintenance.
- Managed build results in Jenkins and deployed workflows for various projects, ensuring smooth software releases.
- Designed and implemented fully automated server build, management, monitoring, and deployment solutions using DevOps technologies like Ansible.
- Worked with Source Code Management (SCM) tools like Git and GitHub for version control and code collaboration.
- Migrated code from Bitbucket to streamline CI/CD pipeline deployment processes.
- Worked on Docker containerization, maintaining Docker images and containers to streamline application deployment.
- Created custom YAML files for Kubernetes objects and workflows, packaging them using Helm charts.
- Performed Linux server administration including patching, user management, log analysis, file system tuning, and performance optimization.
- Automated deployment utilities using Python and PowerShell, integrating Terraform with Jenkins and other CI/CD pipelines.
- Created Chef cookbooks and wrote Ruby recipes to install and configure infrastructure across environments, automating processes with Python scripts.

Environment: Python, Ruby, Chef, PowerShell, Terraform, Jenkins, CI/CD pipelines, YAML, Helm charts, Perl, Bash, Tomcat, Maven, IaC, Ansible, Docker, Kubernetes, Azure Blob Storage, Azure Files, Prometheus, Grafana, New Relic, Azure Log Analytics, Azure Functions, Azure Service Bus, Azure Key Vault, Azure AD, Azure API Management, Azure Storage, Azure SQL/DB, Cosmos DB, Git, GitHub, Splunk, AKS, Datadog, SonarQube, Bitbucket, Docker Images.

Client: Cipla USA, Road Island, NY.

Jan 2016 – Jul 2018

Role: AWS Engineer

Responsibilities:

- Designed, configured, and managed public and private cloud infrastructures on Amazon Web Services (AWS), including EC2, Virtual Private Cloud (VPC), Public and Private Subnets, Security Groups, Route Tables, Elastic Load Balancers (ELB), CloudWatch, and IAM.
- Created AWS launch configurations with customized AMIs to configure auto-scaling groups and implemented AWS solutions using EC2, S3, RDS, DynamoDB, Route 53, ELB, and auto-scaling groups.
- Automated continuous deployment (CD) using Ansible to deploy and configure ELK, and upgraded the entire ELK cluster to newer releases in an OpenStack enterprise environment.
- Configured AWS Multi-Factor Authentication (MFA) in IAM to implement two-step authentication for user access using AWS Virtual MFA, enhancing cloud security with security groups, network ACLs, Internet Gateways, and Elastic IPs.
- Managed local Kubernetes deployments by creating local clusters and deploying application containers to support scalable infrastructure.
- Integrated Jenkins with DevOps tools like Nexus and Ansible, utilizing a CI/CD system on a Kubernetes container environment for building, testing, and deployment workflows.
- Implemented a CI/CD pipeline with version control (Git), Jenkins, Ansible, and Docker to automate the process from code commitment to deployment.
- Designed and maintained Git repositories, providing day-to-day Git support for multiple projects.

- Developed Ansible playbooks for setting up continuous delivery pipelines, provisioning AWS environments, and deploying microservices.
- Utilized NoSQL databases such as Cassandra and MongoDB for scalable database solutions.
- Worked with Kibana and Elasticsearch to identify Kafka message failure scenarios, implementing solutions to reprocess failure messages using Kafka offset IDs.
- Created and maintained the Python deployment scripts for the Web Sphere web application server.
- Involved in designing and deploying a multitude of applications utilizing almost all the AWS stack focusing on high availability, fault tolerance, and auto-scaling in AWS Cloud formation.
- Deployed AWS EC2 instances on platforms such as RHEL, CentOS, Ubuntu (Linux), and Windows to support a variety of application environments.
- Designed, implemented, and maintained infrastructure as code (IaC) using Terraform to automate cloud resource provisioning, achieving consistent and scalable deployments across AWS environments.
- Created Ansible playbook and deployed on nodes, Expert in creating core concepts like Inventory, Playbooks, Roles, Tasks, and Modules.
- Deployed Ansible and Jenkins to complete provision and manage AWS EC2 instances, volumes, DNS, and S3.

Environment: Python, Jenkins, AWS, Ec2, VPC, ELB, CloudWatch, IAM, AMIs, S2, RDS, DynamoDB, Route 53, Auto-Scaling groups, MFA, Kubernetes, Nexus, Ansible, CI/CD, Kibana, Git, Docker, NoSQL, Cassandra, MongoDB, Kafka, Ubuntu, Windows, IaC, Terraform, DNS.

Client: GKG Technologies, Hyderabad, India.

May 2014 – Dec 2014

Role: Build and Release Engineer

Responsibilities:

- Provided Release Engineering services to support engineering plans and schedules, including building, deploying, developing scripts, overseeing branch and merge strategies, and creating automated tools to enhance services for the engineering team.
- Automated daily processes using the continuous integration tool Jenkins, ensuring efficient workflows.
- Migrated J2EE applications to UNIX and Windows environments by configuring applications via Subversion and Jenkins.
- Updated Maven scripts to utilize Artifactory repositories instead of local repositories, optimizing build processes.
- Managed source code in Subversion, including version control, branch creation, and conflict resolution.
- Administered Git version control system, performing tasks such as creating branches, tagging, merging, and resolving conflicts.
- Collaborated with Web Administrators to design and implement technical architecture for websites using Apache, JBoss, and WebLogic, and deployed J2EE applications to these environments.
- Developed and maintained build scripts using ANT (build.xml) and Maven (pom.xml) for efficient build and deployment processes.
- Automated weekly deployments by leveraging Jenkins CI tools, reducing manual intervention and deployment time.
- Demonstrated a solid understanding of monitoring tools like Splunk, New Relic, and Sansui to ensure system reliability and performance.
- Administered JIRA, maintained documentation, and trained users for effective project tracking and management.

Environment: J2EE, Jenkins, UNIX, Windows, Maven, Artifactory, JBoss, Perl Scripts, Shell Scripts, Bash Scripting, Nexus, Jira, Apache Tomcat.

Education:

- Master of Science in Computer and Information Science from Southern Arkansas University, Magnolia, AR.
 - Bachelor of Technology in Information Technology from JNTU Kakinada, India.
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